

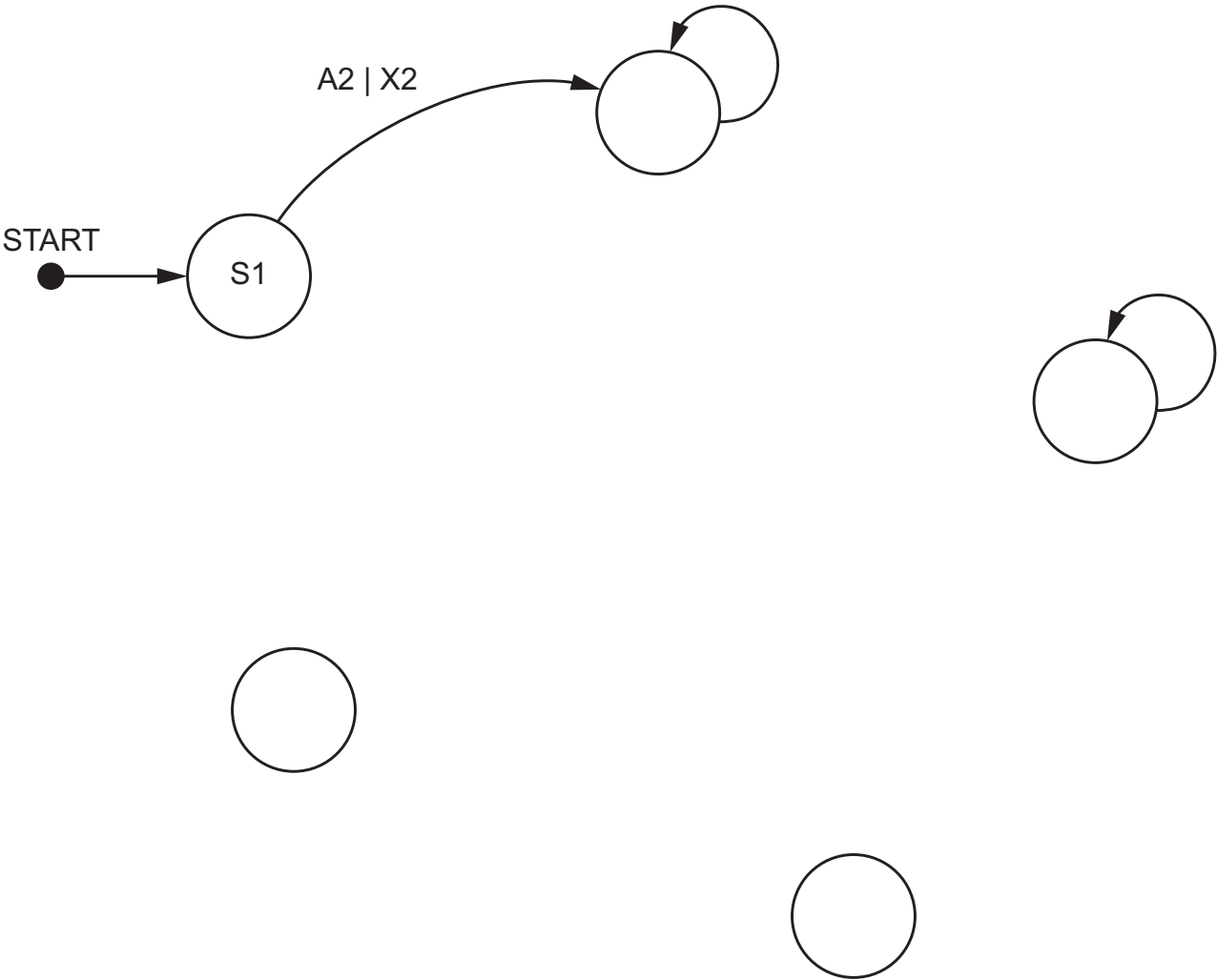
7 There are several different ways to express an algorithm during the design of a program.

- (a) One part of the program contains an algorithm which is represented by a state-transition diagram.

The table shows the inputs, outputs and states for the algorithm:

| Current state | Input | Output | Next state |
|---------------|-------|--------|------------|
| S1            | A2    | X2     | S3         |
| S1            | A1    | X1     | S2         |
| S2            | A4    | X4     | S5         |
| S3            | A1    |        | S3         |
| S3            | A3    | X3     | S2         |
| S3            | A2    | X4     | S4         |
| S4            | A1    | X1     | S4         |
| S4            | A3    |        | S2         |
| S4            | A4    | X4     | S5         |

Complete the state-transition diagram to represent the information given in the table.



[5]

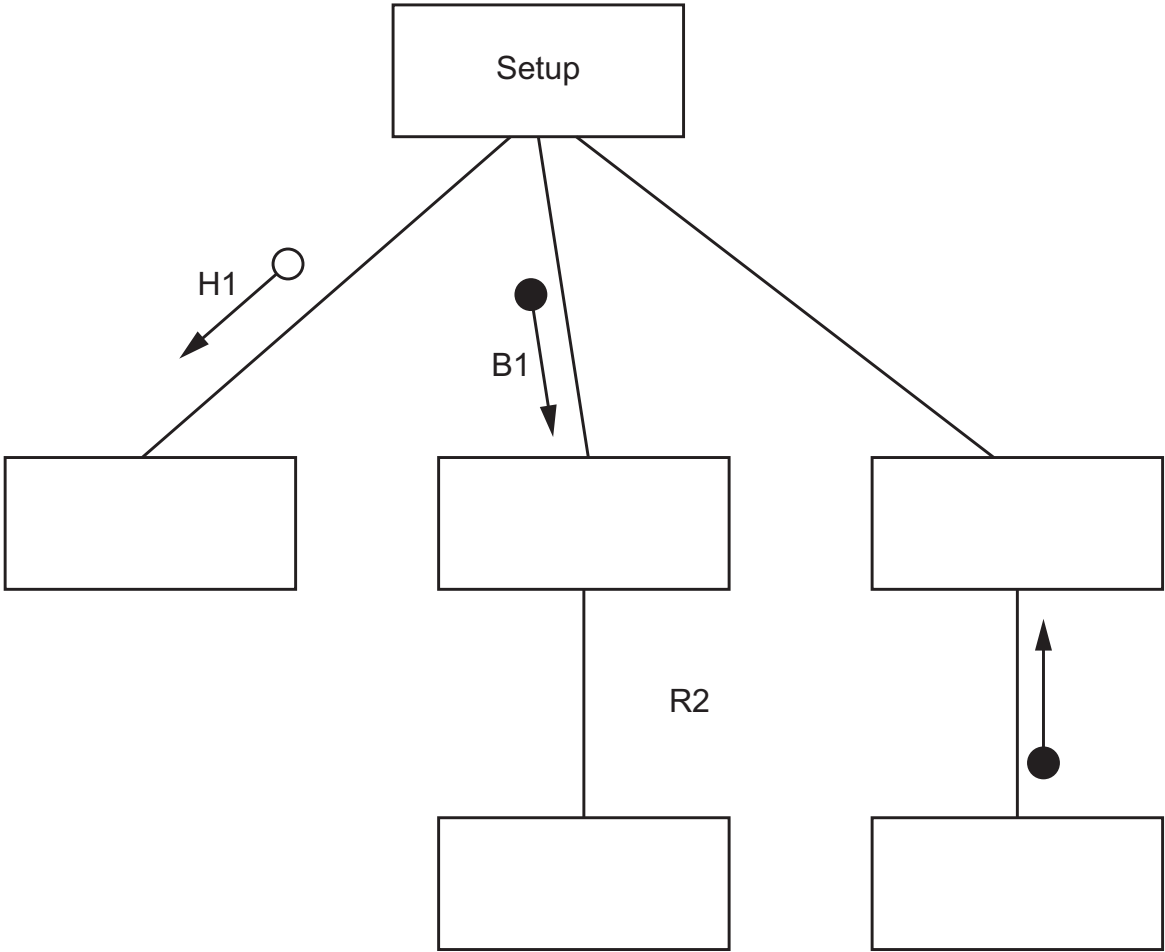
- (b) A structure chart is used to document a different part of the program.

This part of the program contains six modules:

| Pseudocode module header                      |
|-----------------------------------------------|
| PROCEDURE Setup()                             |
| PROCEDURE Restart(H1 : STRING, C1 : INTEGER)  |
| FUNCTION Modify(B1 : BOOLEAN) RETURNS INTEGER |
| PROCEDURE Final(T1 : INTEGER)                 |
| PROCEDURE Update(BYREF R2 : STRING)           |
| FUNCTION Confirm() RETURNS BOOLEAN            |

Module Setup will repeatedly call **three** of the modules.

Complete the structure chart to document the information given in the table.



[5]